VU THI HAI YEN

haiyen96.hp@gmail.com◇ Paris, France ◇ GitHub ◇ LinkedIn

ABOUT ME

Machine Learning Engineer | Applied Scientist | Data Scientist

Machine learning practitioner (current PhD Candidate) with 4+ years across industry and research, taking models from raw, messy data through to deployed services. My work spans graph and neuro-symbolic learning, learning from partially-labelled data, NLP, and mineral prospectivity mapping. Two blind peer-reviewed publications (ACM SAC 2025, ECML PKDD 2026), each beating state-of-the-art baselines. Strong Python/PyTorch; used to collaborating with non-ML domain experts.

WORK EXPERIENCE

Université d'Orléans / BRGM — Orléans, France
PhD Researcher, Machine Learning — *October 2023 - Present*

- Designed *NeuralAdjust*, a **neuro-symbolic PU-learning** method (**NeuPSL, PyTorch**) that dynamically reweights logical rules to resolve conflicting constraints; outperformed the state-of-the-art **graph-based PU method** (GPL) across benchmarks - e.g. F1 **88.1 vs 81.9** on Cora and **73.6 vs 36.2** on Chameleon, while improving class-prior estimation.
  Accepted at ECML PKDD 2026.
- Developed a feature-mining graph-construction method for **mineral prospectivity mapping (MPM)** on an imbalanced dataset (105 positives among 14,218 nodes, 118 features); *increased* recall@top-5% by **12.5%** (GCN) and **9%** (GAT) over the spatial-only baseline, beating **XGBoost** and MLP.
  Published at ACM SAC 2025.
- Processed heterogeneous **Geographic Information System (GIS)** datasets with multiple layers at varying spatial resolutions.
- Translated geoscientists' domain expertise into model constraints and features; delivered **two blind peer-reviewed papers** and presented results to non-ML stakeholders across **8+ talks and seminars**.

ECMWF Code4Earth — Remote
ML Engineer – 5-person team — *2024 - 2025*

- Collaborated in a **team of 5**, mentored by ECMWF scientists, to deliver two open-science ML challenges from prototype to working application - coordinating across data pipeline, modelling, and deployment via shared Git workflows.
- **DSS ML-Costing** (2025): co-built and deployed a cost-estimation service for ECMWF's Data Store - a FastAPI predicting request processing time. Trained **XGBoost/LightGBM** with **Optuna** + 5-fold CV across 18 datasets (up to **866K** samples), reaching $R^2 \approx 0.96$ and MAPE ≈13% on the strongest datasets; contributed schema validation, feature engineering, and model evaluation.
- **KGB-TruthGuiding** (2024): co-built a **GraphRAG** system to ground and visualize LLM answers over ECMWF scientific literature - LLM-based entity/relation extraction (**GPT, Llama 3**) into a knowledge graph, community detection with local/global retrievers; the team assembled a ∼490-document corpus via a custom crawler and **PyMuPDF**.

LIGM Lab, Gustave Eiffel University — Paris, France
Research Engineer — *April 2022 - July 2022*

- Integrated feature vectors of words into **Unitex/GramLab**, an open-source NLP platform, implementing word-tag selection persistence across the C++ core engine and Java visual IDE.
- Built an English word-lattice dataset and a PyTorch model for lattice-structured NLP data.

SYSTRAN — Paris, France
NLP Research Intern — *June 2020 - November 2020*

- Developed contextualized paraphrase generation with tag-specific **Transformer/GPT2** models
- Used the synthetic data to augment **OpenNMT** translation across English-French, English-German, and English-Vietnamese.

EDUCATION

Université d'Orléans *2023 - Present*
- PhD Candidate in Machine Learning, in collaboration with BRGM

École Polytechnique *2019 - 2021*
- Master 2 in Data Science; graduated with High Honors

Université Paris-Est Créteil *2018 - 2019*
- Master 1 in Mathematics and Applications, ranked 2/20

Hanoi National University of Education *2014 - 2018*
- Bachelor of Mathematics; Very Good Degree

PUBLICATIONS

Positive-Unlabeled Learning with Knowledge-Structured Data [code]
Thi Hai Yen VU, Vincent NGUYEN, Christel VRAIN, Hugo BREUILLARD, and Thi Bich Hanh DAO.
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases - ECML PKDD 2026 (acceptance rate 24%)

Features Leverage in Graph Models for Mineral Prospectivity Mapping [link]
Thi Hai Yen Vu, Thi Bich Hanh Dao, Vincent Nguyen, Christel Vrain, and Hugo Breuillard.
Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing, pp. 597-604 (acceptance rate 25%)

ACADEMIC ACTIVITIES

Teaching - Université d'Orléans, bachelor level: Programming and Algorithms (16h practicals); Discrete Mathematics (18h tutorials + 12h practicals).
Talks & Presentations - ANR IA.iO Day; BRGM "Digital Data" seminar (poster); ARETHUSE webinar; BRGM Young Researchers AI Group; MaDiTraCe Workshop; CA team seminar. *2024 - 2025*
Scientific Training - Nordic Probabilistic AI School; Eastern European Machine Learning Summer School; MaDICS Symposium; CNRS Fête de la Science (2024).

LANGUAGES & SKILLS

Programming	Python, Java, C++, R, MATLAB
ML / Deep Learning	PyTorch, PyTorch Geometric, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM
ML Methods	PU & semi/self-supervised learning, graph learning (GCN/GAT), neuro-symbolic (NeuPSL/PSL), clustering, NLP, LLMs & GraphRAG
Data & Libraries	pandas, NumPy, NetworkX, geopandas, PyMuPDF, OpenNMT
Natural Languages	Vietnamese (native), English (proficient), French (upper-intermediate / B2)

ACADEMIC HONORS

Eiffel Excellence PhD Scholarship - French Ministry for Europe and Foreign Affairs *2022*
Masters Scholarship - Foundation of France *2018 - 2020*
Dam Phu My Scholarship for Excellent Students; NKTP Global Award *2016*
NKTP Global Award *2014*